

# CISII Final Report: Vision-Based Localization for Micro-Continuum Surgical Robots: A Sim-to-Real Benchmark and Adaptation Study

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## I. BACKGROUND

Micro-continuum surgical robots have emerged as a promising solution for highly confined anatomical workspaces [1], with applications spanning retinal micro-surgery [2], neurovascular intervention [3], and endoscopic surgery [4]. Despite their dexterity advantages, these systems share a fundamental control challenge: nonlinear hysteresis arising from tendon slack, internal friction, and material compliance complicates precise tip positioning [5].

Accurate Micro-continuum surgical robots localization is therefore essential for enabling automation and ensuring safe operation in microsurgical environments. Tasks such as membrane peeling or catheter navigation require precise knowledge of the instrument tip relative to delicate anatomical structures, where even sub-millimeter errors can cause tissue damage [6]. Reliable localization is thus a key requirement for both safe teleoperation and future autonomous surgical behaviors.

Vision-based localization offers an external feedback mechanism that bypasses explicit hysteresis modeling by operating directly on observed motion outcomes. This project therefore studies how to make vision-based localization practical for data-scarce intraocular robotic settings.

This section situates the project within prior work on micro-continuum surgical robots, simulation environments, and learning-based localization methods. It is intended to provide the broader technical context behind the specific problem defined above.

### A. Micro-Continuum Surgical Robots

Micro-continuum robots have emerged as a compelling solution for surgical tasks requiring access to confined anatomical workspaces with high dexterity. Unlike rigid-link manipulators, continuum robots achieve motion through elastic deformation or tendon-driven actuation, enabling enhanced dexterity and compact profiles suitable for minimally invasive procedures [5].

In retinal and intraocular surgery, the confined geometry of the eye and the fragility of retinal tissue place extreme demands on positioning accuracy. Systems such as the Base Robot (BR) [7] address tremor suppression for rigid instruments, while cable-driven continuum designs extend dexterity to access intraocular structures through small incisions [2]. Concentric tube robots represent another prominent continuum paradigm, with applications spanning minimally invasive surgical manipulation and deep-access interventions

[1], sharing the fundamental challenges of hysteresis and compliance present in cable-driven designs. In endoscopic [8] and catheter-based intervention, similar continuum architectures have been explored extensively, with hysteresis compensation remaining an open control problem across all these systems [3].

A common thread across all these platforms is that kinematic modeling alone is insufficient for precise tip localization under hysteresis [9], motivating external sensing and vision-based feedback as a complementary approach [5]. Nevertheless, vision-based localization in these settings is complicated by the confined, texturally complex workspaces in addition to the lack of large-scale annotated datasets, yet these challenges are directly addressed by our benchmark.

### B. Simulation Frameworks for Surgical Robotics

Physics-based simulation has become an essential tool for surgical robotics research, enabling scalable data generation, policy training, and system evaluation without the cost and risk of real hardware operation. Early simulation environments such as the dVRK simulator [10] focused primarily on replicating the kinematics of commercial surgical platforms for teleoperation research.

More recent frameworks have emphasized learning-oriented design. SurRoL [11] provides a reinforcement learning-compatible environment for the da Vinci Research Kit with task-relevant surgical subtasks. ORBIT-Surgical [12] extends this paradigm using GPU-parallelized simulation to support both reinforcement and imitation learning at scale for augmented surgical dexterity. FF-SRL [13] demonstrates that GPU-accelerated deformable tissue simulation can reduce RL training time by an order of magnitude compared to CPU-based alternatives.

For vision-oriented data generation specifically, the Asynchronous Multi-Body Framework (AMBF) [14] provides real-time physics simulation with native support for synchronized RGB-D capture, segmentation, and pose annotation, making it particularly suitable for intraocular robotic scenarios [15]. Building on AMBF, prior work developed a modular synthetic data generation toolkit for the localization of a custom-built micro-continuum surgical robot (CMSR), demonstrating sub-millimeter pose tracking with simulated depth and zero-shot sim-to-real detection [15]. However, that work evaluates only two vision methods and does not study the effect of incorporation real data. Our work extends this line by providing the first systematic benchmark of vision-

based localization methods across both simulated and real intraocular data. In addition, we characterize how synthetic pretraining interacts with limited quantities of manually annotated real images.

### C. Localization Methods

Vision-based localization for surgical robots spans a broad range of sensing modalities and learning paradigms. We group the 13 evaluated image-based methods into the following categories:

- **Classical baselines** are low-capacity, non-deep-learning models including Template Matching [16], and Color Segmentation [17].
- **Heatmap-based methods** are models that predict dense keypoint likelihood maps followed by soft-argmax coordinate extraction, including SimpleBaseline [18], HR-Net [19], Stacked Hourglass [20], CenterNet [21], and U-Net [22].
- **Direct regression methods** are models that regress coordinates directly from global image features, including ResNet Direct [23] and ViT Direct [24].
- **Integral regression methods** are hybrid approaches that bridge heatmap prediction and coordinate regression through differentiable expectation operators, including Integral Regression [25], ViTPose [26], and DINOv2 Regression [27].
- **Detection-based localization** is a transformer-style detection framework applied to keypoint localization, represented by DETR [28].

## II. PROBLEM

The purpose of this work is to evaluate the efficacy of the state-of-the-art vision-based localization methods for micro-continuum robots. On the data side, acquiring synchronized RGB images with ground-truth keypoint annotations from micro-continuum surgical robots under real intraocular procedures is impractical at scale. As a solution, simulation can help mitigate data shortages by providing synthetic data. To this end, prior work introduced a physics based toolkit for synthetic data generation [15]. In particular, this work demonstrated that the sim-to-real gap in lighting, tissue appearance, and depth availability substantially degrades localization performance on real microscope imagery. Hence our work emerges as a systematic study on how different model families respond to this gap, or how efficiently real annotations finetuning can close it.

## III. APPROACH

### A. Overview

In this work, we address these challenges through three contributions spanning dataset construction, systematic evaluation, and domain adaptation for vision-based localization of micro-continuum surgical robots:

- **Synthetic and real dataset construction:** An automated pipeline built on the AMBF intraocular environment [15] generates annotated RGB frames for vision-based localization. 2D keypoints are derived from

simulator  $(x, y, z)$  states through calibrated camera projection. The synthetic dataset spans 12 controlled visual conditions, combining 4 retinal background textures with 3 lighting configurations.

In addition, we introduce a manually annotated real intraocular RGB dataset of a micro-continuum surgical robot operating inside real and phantom porcine eyes under a surgical microscope.

- **Multi-method benchmark:** A systematic evaluation of 13 vision-based localization methods spanning classical, heatmap, regression, and detection paradigms, benchmarked on both simulated and real intraocular data under zero-shot sim transfer and real-only training regimes, establishing the first comprehensive performance baseline for this domain
- **Domain adaptation study:** A sim-to-real finetuning analysis across model families and numbers of manually annotated real images, demonstrating that synthetic pretraining provides a complementary axis of improvement beyond the saturation point of real data alone, with practical guidelines for data-efficient deployment

### B. Simulation Toolkit Design

The simulation environment is built on the Asynchronous Multi-Body Framework (AMBF) [15], which provides real-time physics simulation with tendon-driven actuation, synchronized RGB-D capture, and native support for external control interfaces. The environment comprises three modular components — the robot model, the eyeball model, and the virtual camera/microscope — each defined by AMBF Description Format (ADF) files that specify geometry, inertial properties, and joint structure independently, allowing substitution of any component without modifying the others.

**Robot model.** The eye snake is modeled as a chain of 11 disc-like segments connected by alternating revolute joints, replicating the physical cable-driven continuum structure. A two fingers grip is attached: one fixed and one movable, consistent with the real CMSR hardware. Geometry and inertial parameters are matched to the physical robot’s dimensions and fabricated segment masses, ensuring that the simulated kinematics reproduce the real system’s bending behavior under actuation.

**Eyeball model.** A 12 mm-radius spherical eyeball is constructed in Blender<sup>1</sup> and exported to ADF format. The inner surface accepts interchangeable retinal texture maps derived from real porcine intraocular images, providing anatomically grounded background appearance while allowing controlled texture variation across data collection runs.

**Camera model.** A virtual camera is mounted above the eyeball to replicate the viewpoint of a Leica surgical microscope (Leica Microsystems, Germany). Intrinsic parameters — focal length, principal point, and lens distortion coefficients — are obtained via standard checkerboard calibration of the physical microscope and transferred directly to the virtual camera, ensuring that simulated images share the

<sup>1</sup><https://www.blender.org/>

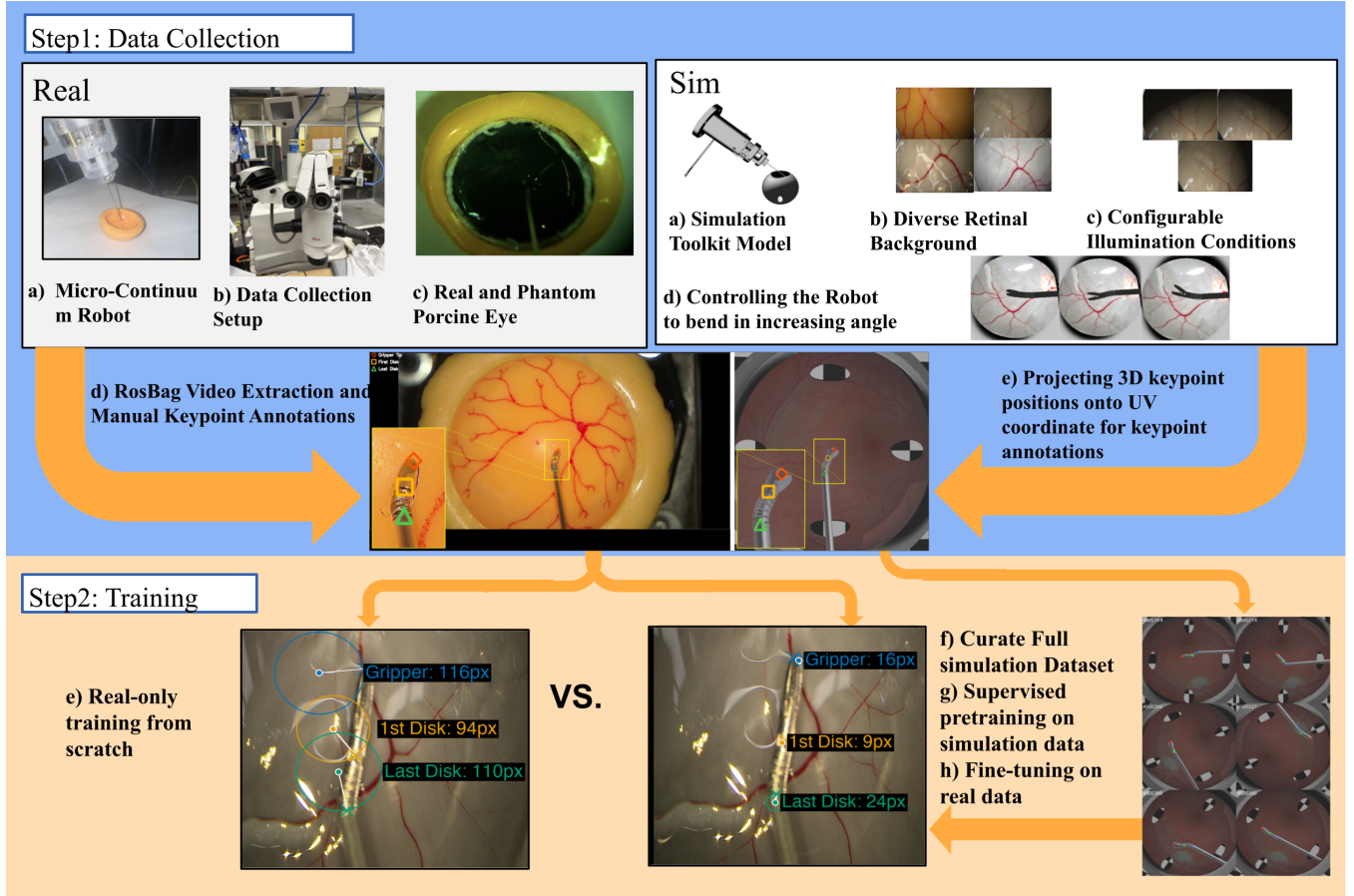


Fig. 1. Overview of the dual-domain data collection and training pipeline. **Step 1 – Data Collection.** *Real* (left): the micro-continuum surgical robot is mounted on a BR base under a Leica surgical microscope and operated inside real and phantom porcine eyes; RGB video is extracted from ROS bags and annotated with three manual keypoints. *Simulation* (right): an AMBF-based toolkit models the robot geometry with configurable retinal background textures and illumination conditions; the robot is programmatically controlled through increasing bending angles, and 3D keypoint positions are projected onto UV image coordinates for automatic annotation. **Step 2 – Training.** Real-only training from scratch (left) is compared against sim-to-real finetuning (right), where models are first pretrained on the curated simulation dataset (f–g) then finetuned on real data (h), yielding substantially lower localization error across all three keypoints.

same projection geometry as real captures. A distortion plugin applies the calibrated radial and tangential distortion coefficients at render time.

**Robot control.** Rather than modeling the full multi-stage hysteresis of the physical system, the control layer employs a simplified constant-curvature kinematic formulation. The entire bending section is approximated as a circular arc parameterized by total pitch and yaw angles  $(\phi_{\text{total}}, \psi_{\text{total}})$ , with curvature vector:

$$\vec{\kappa} = \left[ \frac{\phi_{\text{total}}}{L}, \frac{\psi_{\text{total}}}{L}, 0 \right]^T, \quad \theta = \|\vec{\kappa}\|L \quad (1)$$

where  $L$  is the arc length. The resulting tip pose in the world frame is computed via the Rodriguez rotation formula [15] and sent to the simulation. Bending angles are constrained to  $(-20^\circ, 20^\circ)$  in both pitch and yaw to match the physical operating envelope. Forward and inverse kinematics with a remote-center-of-motion (RCM) constraint are implemented in Python, enabling efficient posture sampling for large-scale data generation without per-step manual control.

### C. Simulation Data collection

Using the toolkit described above, we generate a dataset by sampling 300 robot configurations within the reachable workspace and rendering each configuration under controlled visual variations.

Visual diversity is introduced along two controlled axes. Lighting is varied across three discrete configurations spanning front illumination to backlighting, modulating overall illumination levels and shadow distributions. Four distinct retinal background textures derived from real porcine eye images are applied to the eyeball inner surface, reducing overfitting to a single anatomical appearance. The combination yields 12 distinct visual configurations per robot pose, as illustrated in Fig. 1.

For each rendered frame, the 2D image-space coordinates  $(u, v)$  of each keypoint are obtained by projecting the simulator’s ground-truth 3D keypoint positions  $\mathbf{p}_{\text{target}}^{\text{world}}$  into the camera frame via:

$$(u, v)^T = \mathbf{K} \cdot \mathbf{P} \cdot (\mathbf{T}_{\text{camera}}^{\text{world}})^{-1} \cdot \mathbf{p}_{\text{target}}^{\text{world}} \quad (2)$$

where  $\mathbf{K}$  is the camera intrinsic matrix,  $\mathbf{P}$  is the projection

matrix, and  $\mathbf{T}_{\text{camera}}^{\text{world}}$  is the camera pose in world coordinates [15]. Labels are generated for three keypoints — gripper tip, first disk (segment 11), and last disk (segment 1) — matching the keypoint definitions used in real data collection to enable direct cross-domain comparison without annotation format discrepancy.

#### D. Real Data Collection

Real data is collected using a micro-continuum surgical robot mounted on a BR base [7], which provides stable positioning of the overall surgical apparatus. The continuum segment is inserted into a physical eyeball model through a small incision, replicating the intraocular workspace of a clinical retinal procedure. Enucleated porcine eyes are used as the anatomical substrate, providing natural retinal texture, intraocular fluid, and tissue reflectance properties that are not fully replicated in simulation.

A Leica surgical microscope is positioned above the eye and provides the imaging viewpoint for all real data collection, consistent with the camera model used in simulation [15]. RGB video is recorded directly from the microscope output at  $1920 \times 1080$  resolution at a fixed focal plane. The robot is manually actuated through a range of pitch and yaw configurations across recording sessions to capture diverse bending states and visual conditions. No depth sensor is used, as accurate depth acquisition under surgical microscope optics at millimeter scales remains an open hardware challenge [15]; ground-truth labels are therefore restricted to 2D image-space keypoint coordinates.

Frames are extracted from each video sequence and annotated using the same three-keypoint protocol as simulation, labeling the gripper tip, first disk (segment 11), and last disk (segment 1) when visible. Frames where any keypoint is occluded or outside the field of view are excluded. All annotations are performed manually using a custom labeling interface and reviewed for consistency.

#### E. Dataset Statistics and Splits

Fig. 2 summarizes the real data domain. The real dataset comprises 6 video sequences (V1–V6) recorded during ex-vivo porcine eye experiments with the CMSR, totaling 400 extracted frames. Of these, 217 frames contain complete 3-keypoint annotations (grripper tip, first disk, last disk) verified as visible and unoccluded; the remaining 183 frames are unlabeled or partially annotated. Fig. 2 illustrates both the spatial distribution of gripper tip positions across videos and the partitioning pipeline.

From the 217 annotated frames, we construct a fixed held-out test set of 50 frames and a training pool of up to 150 frames, with stratified sampling to maintain proportional video representation in both splits. The 17 remaining annotated frames are held in reserve. From the training pool, nested subsets of  $N \in \{30, 60, 90, 120, 150\}$  frames are constructed for data efficiency analysis, with strict inclusion ( $N = 30 \subset N = 60 \subset \dots \subset N = 150$ ) enforced at each scale to ensure that improvements from additional data are not confounded by changes in sample composition. In all

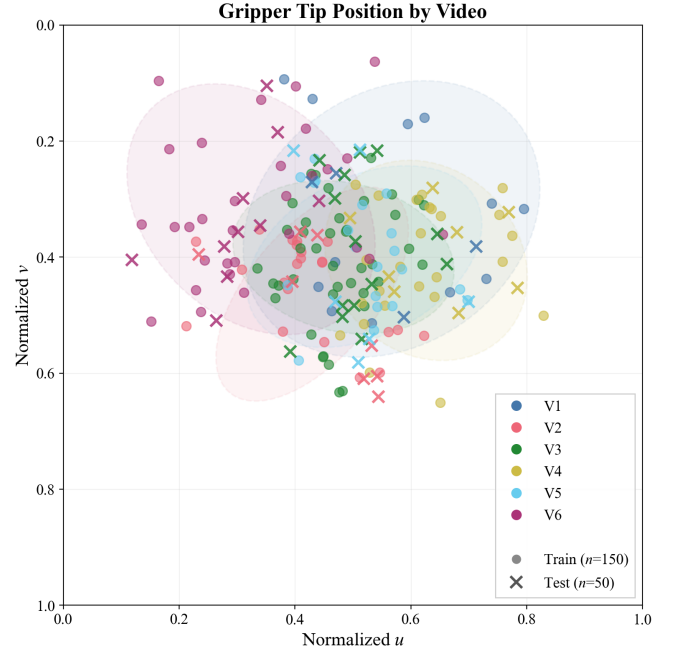


Fig. 2. Gripper tip positions across 150 training (circles) and 50 held-out test (crosses) images. V1–V6 denote six recording sessions with different camera placements and robot poses. Ellipses show per-video  $1\sigma$  covariance. The train/test split preserves spatial coverage across all sessions.

experiments, models are evaluated exclusively on the 50-frame real test set ( $1920 \times 1080$  native resolution), providing a consistent cross-domain evaluation benchmark regardless of training domain or resolution.

The simulation dataset comprises 300 robot configurations rendered across 12 visual conditions (4 retinal textures  $\times$  3 lighting setups), yielding 300 annotated frames with automatically projected 2D gripper labels. These are expanded to 900 samples via domain randomization augmentations (brightness reduction, Gaussian blur, scaling, noise injection).

#### F. Training and Fine-tuning Strategy

We evaluate each approach under two training regimes: *Real-Only*, trained from scratch on  $N$  real images; and *Sim*→*Real Fine-tuning (FT)*, pretrained on 300 simulation images then fine-tuned on  $N$  real images. All models use AdamW [29] with cosine annealing,  $\text{lr} = 10^{-3}$ , and 150 epochs unless otherwise noted.

**Initialization strategy.** A critical finding is that naively loading all simulation-pretrained weights for fine-tuning can *degrade* performance below the real-only baseline when real data is scarce. We trace this to a backbone corruption effect: simulation pretraining overwrites general-purpose ImageNet [30] features with simulation-specific representations that cannot be recovered from  $\leq 150$  real samples.

This motivates a backbone-dependent initialization strategy:

- *Approaches with ImageNet-pretrained backbones* (Integral Regression, SimpleBaseline, ResNet Direct, CenterNet, DETR, Image UV+3D): we apply **hybrid**

**initialization**—retain the original ImageNet backbone weights and load only the sim-pretrained task heads (deconvolution layers, regression heads, or decoders). This preserves the general visual features while benefiting from sim-learned task structure.

- *Approaches without pretrained backbones* (Hourglass, HRNet, U-Net, ViT Direct, ViTPose): since no general-purpose features exist to preserve, we load all simulation-pretrained weights and fine-tune the entire network.
- *DINOv2 Regression*: the self-supervised DINOv2 backbone is kept frozen throughout both pretraining and fine-tuning; only the lightweight regression head is updated.

**Multi-resolution ensemble.** To exploit the complementary strengths of different input resolutions, we train Integral Regression [25] models independently at  $512 \times 512$ ,  $768 \times 768$ , and  $1024 \times 1024$  pixels. At inference, each model runs at 5 scales (training resolution  $\pm 32$  and  $\pm 64$  px), and all  $K = 3 \times 5 = 15$  predictions are averaged in normalized  $[0, 1]$  coordinate space:

$$(\hat{u}, \hat{v}) = \frac{1}{15} \sum_{k=1}^3 \sum_{s \in \mathcal{S}_k} f_k^{(s)}(I) \quad (3)$$

where  $f_k^{(s)}$  denotes model  $k$  evaluated at scale  $s$ .

## IV. RESULTS

### A. Training Protocol

We study two input resolutions:

**224px.** Models train for 30 epochs with batch size 8. Real-only training uses  $\text{lr} = 10^{-3}$ ; finetuning uses  $\text{lr} = 10^{-4}$  (lower rate to preserve pretrained features). Data efficiency is evaluated over nested subsets  $N \in \{30, 60, 90, 120, 150\}$  (Table I).

**512px.** Models train for 200 epochs with batch size 4,  $\text{lr} = 10^{-4}$ , and a 5-epoch linear warmup. The longer schedule compensates for the smaller effective batch at higher resolution. All models use  $N = 150$  real images (Table II). We additionally evaluate a *multi-resolution ensemble* that averages normalized predictions from Integral Regression models trained independently at 512, 768, and 1024 px.

**Unified protocol.** To ensure fair comparison across architectures, all deep learning approaches share the same hyperparameters within each resolution-regime setting. We deliberately avoid per-model tuning: our goal is to evaluate architectural inductive biases under controlled conditions, not optimized configurations. This may disadvantage architectures that require specialized schedules (e.g., vision transformers typically benefit from longer warmup and lower learning rates). The two classical baselines (Template Matching and Color Segmentation) use their own fitting procedures and are unaffected by these hyperparameters.

### B. Metrics

**Mean pixel error (px).** We report the Euclidean distance between predicted and ground-truth keypoints in pixel space,

$e = \sqrt{(\hat{u} - u)^2 + (\hat{v} - v)^2}$ , where  $(u, v)$  are pixel coordinates on a  $1920 \times 1080$  image. Models predict normalized coordinates in  $[0, 1]$ , which are scaled back by image dimensions for error computation. Results are reported per keypoint (gripper, first\_disk, last\_disk) and overall (mean across all three keypoints over all test samples).

**Why pixel error (not mm).** Real data provides only 2D image-space labels, so pixel error is the most direct and consistent metric. Because models predict in normalized  $[0, 1]$  space, pixel error is resolution-independent. For reference, at a typical working distance of a CMSR,  $1 \text{ px} \approx 0.015 \text{ mm}$  (so  $21.5 \text{ px} \approx 0.32 \text{ mm}$ ).

**Test protocol.** We evaluate on 50 held-out real frames from 6 video sequences at  $1920 \times 1080$  resolution. There is zero overlap between train and test frames (verified by frame index). Each model is evaluated on the same fixed test set across all experiments, and all reported numbers are from this test set (not validation).

““

### C. Results and Analysis

Tables I and II present the full benchmark. We organize our analysis into five findings ordered by practical significance.

**Finding 1: Sim-to-real finetuning is effective at 512px but unreliable at 224px.** At 512px (Table II), 6 of 9 deep learning approaches improve with sim pretraining and finetuning. Integral Regression shows the largest gain:  $40.7 \rightarrow 21.5 \text{ px}$  ( $-47\%$ ), followed by DETR from  $95.9$  to  $76.9 \text{ px}$  ( $-20\%$ ) and Hourglass from  $70.2$  to  $60.0 \text{ px}$  ( $-15\%$ ). Three approaches show slight degradation: CenterNet ( $+5\%$ ), DINOv2 ( $+3\%$ ), and U-Net ( $+1\%$ )—all approaches that either already achieve low real-only error or use a frozen pretrained backbone.

At 224px with  $N = 150$  (Table I), the pattern reverses: 6 of 11 approaches *worsen* with finetuning. The top-performing approaches degrade most—Hourglass rises from  $62.3$  to  $74.2 \text{ px}$  ( $+19\%$ ) and ResNet Direct from  $68.2$  to  $82.3 \text{ px}$  ( $+21\%$ ). Only approaches with high real-only error benefit: DETR improves from  $212.8$  to  $112.8 \text{ px}$  ( $-47\%$ ) and SimpleBaseline from  $182.6$  to  $120.0 \text{ px}$  ( $-34\%$ ). At low resolution, the information bottleneck limits how well simulation features can transfer, causing negative transfer for approaches that already fit the real distribution.

**Finding 2: Finetuning corrects harder samples & rescues challenging cases** Per-sample analysis reveals that sim-to-real preferentially fixes the samples where real-only training fails most (Fig. 3). For Integral Regression at 512px, finetuning **improves 46 of 50** test samples; the hardest quartile (by real-only error) reduces error by  $16.5 \text{ px}$  on average. The correlation between real-only difficulty and finetuning improvement is positive for 6 of 7 models. Analysis of ground-truth keypoint positions shows that hard samples exhibit  $3\times$  higher positional variance and  $1.8\times$  larger keypoint spread ( $225$  vs.  $127 \text{ px}$ )—these are extended or unusual poses that the 150 real training images undersample. Simulation



TABLE I

DATA EFFICIENCY AT  $224 \times 224$  PX: MEAN PIXEL ERROR (PX) ON 50 HELD-OUT REAL TEST IMAGES ( $1920 \times 1080$ ). REAL-ONLY: TRAINED FROM SCRATCH ON  $N$  REAL IMAGES ( $LR=10^{-3}$ ). FT: SIM-PRETRAINED (300 SIM IMAGES) THEN FINETUNED ON  $N$  REAL IMAGES ( $LR=10^{-4}$ ). ALL RUNS USE 30 EPOCHS, BATCH SIZE 8, COSINE SCHEDULER. BEST RESULT PER APPROACH IN **BOLD**.

| Approach            | Category   | Backbone    | Params | Real-Only (224px) |        |        |              |              | Sim→Real Finetuning (224px) |        |        |              |              |
|---------------------|------------|-------------|--------|-------------------|--------|--------|--------------|--------------|-----------------------------|--------|--------|--------------|--------------|
|                     |            |             |        | $N=30$            | $N=60$ | $N=90$ | $N=120$      | $N=150$      | $N=30$                      | $N=60$ | $N=90$ | $N=120$      | $N=150$      |
| Hourglass           | Heatmap    | Hourglass-2 | 6.6M   | 210.7             | 226.3  | 153.4  | 81.5         | <b>62.3</b>  | 210.0                       | 128.3  | 108.0  | 95.7         | 74.2         |
| ResNet Direct       | Regression | ResNet-50   | 24.3M  | 147.3             | 248.4  | 137.0  | 73.5         | <b>68.2</b>  | 197.9                       | 161.2  | 118.7  | 81.9         | 82.3         |
| CenterNet           | Heatmap    | ResNet-18   | 14.0M  | 190.9             | 139.5  | 109.5  | 136.0        | <b>78.9</b>  | 152.6                       | 120.9  | 99.9   | 92.8         | 81.7         |
| Integral Regression | Regression | ResNet-50   | 34.0M  | 166.3             | 130.7  | 266.5  | 153.7        | <b>89.6</b>  | 248.4                       | 206.1  | 107.3  | 91.5         | 93.4         |
| SimpleBaseline      | Heatmap    | ResNet-50   | 34.0M  | 208.4             | 585.1  | 199.2  | <b>101.3</b> | 182.6        | 207.6                       | 186.8  | 155.7  | 127.7        | 120.0        |
| DETR                | Detection  | ResNet-50   | 12.2M  | 213.1             | 152.4  | 205.7  | 115.3        | 212.8        | 206.5                       | 164.6  | 135.2  | <b>107.7</b> | 112.8        |
| U-Net               | Heatmap    | U-Net       | 31.0M  | 232.4             | 214.9  | 196.2  | 203.1        | 203.4        | 219.6                       | 202.1  | 186.7  | <b>176.1</b> | 189.9        |
| ViTPose             | Regression | ViT-Small   | 17.2M  | 220.6             | 209.3  | 188.1  | 186.5        | 186.8        | 240.9                       | 222.7  | 194.8  | 188.6        | <b>172.1</b> |
| HRNet               | Heatmap    | HRNet-W32   | 28.5M  | 229.4             | 231.3  | 202.1  | 193.6        | <b>182.4</b> | 231.6                       | 210.5  | 210.5  | 191.6        | 193.9        |
| DINOv2 Regression   | Regression | DINOv2-S    | 13.1M  | 212.1             | 204.8  | 214.3  | <b>183.3</b> | 212.5        | 231.1                       | 220.5  | 207.6  | 201.6        | 186.8        |
| ViT Direct          | Regression | ViT-Small   | 11.2M  | 269.1             | 212.9  | 201.5  | 190.8        | <b>189.4</b> | 261.1                       | 252.2  | 212.2  | 192.5        | 212.6        |
| Color Segmentation  | Classical  | —           | 163    | 694.4             | 821.7  | 396.1  | 581.4        | 305.7        | 312.4                       | 300.5  | 289.2  | <b>281.7</b> | 293.9        |
| Template Matching   | Classical  | —           | 4K     | 443.0             | 251.7  | 282.8  | 242.8        | <b>230.5</b> | —                           | —      | —      | —            | —            |

TABLE II

HIGH-RESOLUTION RESULTS AT  $512 \times 512$  PX WITH  $N=150$  REAL TRAINING IMAGES. MEAN 2D PIXEL ERROR (PX) ON 50 HELD-OUT REAL TEST FRAMES AT  $1920 \times 1080$ . *Real-Only*: TRAINED FROM SCRATCH ON REAL DATA. *Sim→Real FT*: PRETRAINED ON 300 SIMULATION IMAGES THEN FINETUNED ON REAL DATA.  $\Delta\%$ : RELATIVE CHANGE IN ERROR FROM FINETUNING (NEGATIVE = IMPROVED, POSITIVE = DEGRADED). BEST RESULT PER APPROACH IN **BOLD**.

| Approach                                                | Real-Only        | Sim→Real FT      | $\Delta\%$ |
|---------------------------------------------------------|------------------|------------------|------------|
| Integral Reg.                                           | 40.7             | <b>21.5</b>      | −47        |
| DINOv2 Reg.                                             | <b>40.7</b>      | 41.8             | +3         |
| Hourglass                                               | 70.2             | <b>60.0</b>      | −15        |
| SimpleBaseline                                          | 83.3             | <b>72.5</b>      | −13        |
| DETR                                                    | 95.9             | <b>76.9</b>      | −20        |
| ResNet Direct                                           | 90.5             | <b>79.6</b>      | −12        |
| CenterNet                                               | <b>80.6</b>      | 84.7             | +5         |
| U-Net                                                   | <b>83.8</b>      | 84.4             | +1         |
| HRNet                                                   | 152.8            | <b>133.4</b>     | −13        |
| ViT Direct                                              | N/A <sup>b</sup> | N/A <sup>b</sup> | —          |
| ViTPose                                                 | N/A <sup>b</sup> | N/A <sup>b</sup> | —          |
| Template Match.                                         | 358.0            | —                | —          |
| Color Seg.                                              | 841.0            | —                | —          |
| <i>Multi-res. ensemble (IntReg FT):</i><br>512+768+1024 |                  |                  | —          |
|                                                         | —                | <b>16.3</b>      | —          |

<sup>b</sup>ViT positional embeddings incompatible at 512px.

data, which covers comprehensive elevation angles with uniform pose sampling, provides exactly this missing coverage, explaining why pretraining on simulation specifically rescues the challenging cases.

### Finding 3: Data scaling at 224 px is non-monotonic.

Table I reveals that most approaches do not improve monotonically with more training data at 224 px. SimpleBaseline varies from 208 px ( $N=30$ ) to 585 px ( $N=60$ , catastrophic overfitting) to 101 px ( $N=120$ , best) to 183 px ( $N=150$ ). Integral Regression spikes to 267 px at  $N=90$  despite achieving 131 px at  $N=60$ . Only 2 of 11 deep learning ap-

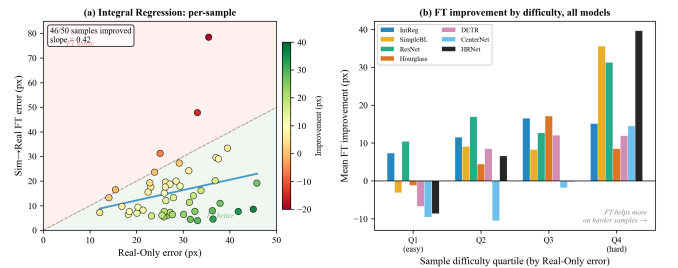


Fig. 3. Per-sample analysis of sim-to-real finetuning benefit at 512 px. (a) Scatter of Real-Only vs. Sim→Real FT error for Integral Regression across all 50 test frames. Points below the dashed diagonal ( $y=x$ ) indicate FT improvement; color encodes absolute improvement in px. **46 of 50 samples improve with finetuning** (slope=0.42), with larger gains on samples where real-only error is highest. (b) Mean FT improvement per difficulty quartile (Q1=easiest, Q4=hardest by real-only error) across 7 models. Finetuning consistently degrades or provides negligible benefit on easy samples (Q1) while strongly helping hard samples (Q4), with improvements from 40 px for HRNet and 32 px for ResNet Direct. This pattern indicates that simulation data compensates for the limited pose coverage of real training data, specifically rescuing underrepresented configurations.

proaches (CenterNet, Hourglass) scale approximately monotonically. Sim pretraining stabilizes this behavior: CenterNet FT decreases monotonically from 153 px ( $N=30$ ) to 82 px ( $N=150$ ), and DETR FT from 207 to 113 px. This suggests simulation pretraining regularizes training even when it does not reduce the final error.

**Finding 4: Resolution provides the largest single improvement.** Comparing best results across resolutions, upgrading from 224 to 512 px yields larger gains than any architecture change at fixed resolution. Integral Regression improves from 89.6 to 40.7 px (−55%), DINOv2 from 183.3 to 40.7 px (−78%), and U-Net from 186.7 to 83.8 px (−55%). At 224 px, the gap between the best deep learning approach (Hourglass, 62.3 px) and worst (ViT Direct, 189.4 px) is  $3.0\times$ ; at 512 px with finetuning, the gap between best (Integral Regression, 21.5 px) and worst (HRNet, 133.4 px) widens to  $6.2\times$ —resolution amplifies rather than levels architectural differences.

### Finding 5: Coordinate decoding determines finetuning

**benefit.** Among heatmap approaches at 512 px, the decoding strategy predicts finetuning gain. Soft-argmax (Integral Regression) computes a differentiable weighted average over the heatmap, enabling end-to-end gradient flow and the largest finetuning improvement (−47%). Grid-quantized argmax approaches (CenterNet) snap predictions to discrete grid locations, disrupting gradient flow from sim-pretrained features and showing no benefit (+5%). Multi-scale deconvolution approaches (SimpleBaseline, Hourglass) achieve moderate finetuning gains (−13% to −15%). Direct regression methods (DINOv2) bypass spatial decoding entirely: DINOv2 matches Integral Regression at 40.7 px real-only but gains nothing from finetuning (+3%), suggesting its frozen backbone already captures sufficient features without adaptation.

**Finding 6: Multi-resolution ensemble provides complementary gains.** Averaging predictions from three Integral Regression models trained at 512, 768, and 1024 px—each with five-scale test-time augmentation (15 total scales)—achieves 16.3 px, a 24% reduction over the best single model (21.5 px). Different resolutions favor different keypoints: 1024 px excels for the small gripper tip, while 512 px better captures the larger disk structures. This ensemble requires no additional training, only inference-time computation.

## V. SIGNIFICANCE

We presented a systematic benchmark of 13 vision-based approaches across five categories—heatmap regression, direct regression, detection, segmentation, and classical methods—for localizing three keypoints on the CMSR from monocular RGB images. Using clean, non-overlapping train/test splits with 30–150 real annotated frames and 300 simulation images, we identified three actionable findings.

First, input resolution is the single most impactful factor: upgrading from  $224 \times 224$  to  $512 \times 512$  px reduces Integral Regression error from 89.6 px to 40.7 px (−55%), a larger improvement than any architecture change or data scaling at 224 px. Seven of thirteen approaches improve at higher resolution, with gains strongest for models with large pretrained backbones (DINOv2 −80%, U-Net −57%). Conversely, classical methods collapse—Template Matching prediction spread drops to 4% of ground-truth variation, effectively outputting a fixed point regardless of input.

Second, simulation pretraining and resolution interact multiplicatively: at 512 px, sim-to-real finetuning reduces Integral Regression error a further 47% ( $40.7 \rightarrow 21.5$  px), whereas finetuning provides no consistent benefit at 224 px. The combination of high resolution and pretraining yields our best single-model result of 21.5 px ( $\approx 0.32$  mm at typical working distance).

Third, multi-resolution ensembling provides complementary gains: averaging predictions from three Integral Regression models trained at 512, 768, and 1024 px with five-scale test-time augmentation each achieves 16.3 px overall error, a 24% reduction over the best single model, as different resolutions capture complementary spatial information (1024

px excels for the small gripper tip; 512 px for the larger disk keypoints).

These results establish a practical recipe for surgical robot keypoint localization with limited annotations: train at the highest feasible resolution using soft-argmax heatmap regression, leverage simulation pretraining for finetuning, and ensemble across resolutions when latency permits.

## VI. MANAGEMENT SUMMARY

### A. Who Did What

This was an individual project. I am responsible for the full technical and documentation workflow, including literature review, experiment planning, synthetic and real data preparation, manual annotation, model training, benchmarking, analysis, figure generation, wiki updates, poster preparation, and final report writing. Mentor feedback supported project scoping, technical direction, and review of intermediate and final results.

### B. Accomplished Versus Planned

The project achieved its main technical objective: developing and systematically evaluating vision-based localization methods for a micro-continuum surgical robot using both synthetic and real intraocular data. The final outcome includes an end-to-end synthetic-data workflow, a manually annotated real dataset, and a benchmark spanning 13 localization methods across real-only, sim-only transfer, and sim-to-real finetuning settings.

Relative to the original plan, the final outcome is stronger in empirical breadth than a minimal prototype. Instead of validating only one or two methods, the final report compares multiple model families across resolutions and data regimes, making the result more useful as a reusable benchmark for future lab work. The project also produced a clearer practical conclusion than originally anticipated: high resolution and simulation pretraining are the two most important factors for strong real-image localization performance.

### C. What Might Be Next

Several next steps would extend this project beyond the current benchmark and move it closer to practical deployment. One immediate direction is to collect additional real stereo microscope video from pig-eye experiments so that future models can exploit binocular visual cues for more accurate and more robust localization. A larger real stereo dataset would also improve coverage of tool poses, tissue appearances, and illumination conditions that remain under-represented in the current monocular benchmark.

Another important direction is sensor fusion. The present study focuses on vision-only localization, but future systems could combine image-based predictions with other modalities such as robot encoder values. Fusing visual measurements with encoder-derived motion information may improve temporal stability, reduce ambiguity during partial occlusion, and provide a more robust estimate when either sensing channel becomes unreliable on its own.

Additional future directions include integrating localization into closed-loop control, evaluating performance on longer surgical sequences, and studying how well the trained models generalize across different eyes, tools, microscopes, and operating conditions.

#### D. What We Learned

This project reinforced several technical and project-management lessons. Technically, the results show that image resolution and pretraining strategy can matter as much as architecture choice, and that simulation is most useful when it expands pose coverage that limited real data does not capture well. The work also highlighted the importance of reproducible training pipelines, careful train/test split design, and consistent evaluation metrics when comparing many methods fairly.

From a project-management standpoint, this effort showed that documentation, experiment tracking, and dataset organization must be treated as first-class deliverables rather than afterthoughts. Maintaining clear records of runs, figures, annotations, and wiki links throughout the semester makes the final report much easier to assemble and improves reproducibility for future students and collaborators.

### VII. TECHNICAL APPENDICES

#### A. Required Documents Summary

The following required documents are available on the project wiki page at:

<https://ciis.lcsr.jhu.edu/doku.php?id=courses:456:2026:projects:456-2026-05:project-05>

All documents are also mirrored on GitHub at:

<https://github.com/ChunTungZhuangLeo/CISII-Documents>

| Document                  | Location               |
|---------------------------|------------------------|
| Functional Specifications | 01-Project-Plan/       |
| Design Specifications     | 01-Project-Plan/       |
| Test Plan                 | 01-Project-Plan/       |
| User Documentation        | 01-Project-Plan/       |
| System Architecture (UML) | 01-Project-Plan/       |
| Background Reading        | 06-Background-Reading/ |
| Checkpoint Presentation   | 07-Project-Checkpoint/ |
| Final Poster              | 09-Final-Presentation/ |
| Final Report              | 10-Final-Report/       |

#### B. Repositories and Data Locations

##### Documentation Repository (Public):

<https://github.com/ChunTungZhuangLeo/CISII-Documents>

Contains: Project proposal, presentations, specification documents, final report, and poster.

##### Code Repository (Private):

Training code, model implementations, and checkpoints will be archived in the official Eye Snake robot private repository

maintained by the LCSR lab. Access can be requested through the project mentors.

##### Real Video Data:

The 6 ex-vivo porcine eye surgery videos (V1–V6) and extracted frames with manual keypoint annotations are stored on the CIS II Microsoft Teams channel under the project folder. Total size: approximately 1.2 GB.

##### Simulation Data:

The 300 AMBF-generated simulation images with automatic 2D keypoint labels are stored alongside the code repository. Retinal texture images and AMBF model files are included.

##### Model Checkpoints:

Trained model weights for all 13 approaches at 224px and 512px resolutions (~2 GB total) are archived with the code repository for reproducibility.

#### C. Archival and Accessibility Notes

- **Wiki:** The project wiki page will remain accessible to JHU students and faculty indefinitely. All links to external documents point to the public GitHub repository.
- **GitHub:** The CISII-Documents repository is public and will remain available. Large binary files (videos, checkpoints) are stored using Git LFS or external hosting due to GitHub size limits.
- **Teams/SharePoint:** Video data on the CIS II Teams channel is accessible to course participants. For long-term archival, copies have been provided to the faculty mentor.
- **Code Handoff:** Source code and trained models will be transferred to the lab’s internal repository to ensure continuity for future students working on the Eye Snake project.
- **Reproducibility:** The GitHub repository includes a `requirements.txt` and training configuration files to enable reproduction of all benchmark results reported in this work.

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